

SATVIK PRASAD

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EDUCATION

Georgia Institute of Technology

B.S. in Computer Engineering, Minor in Mathematics

- John Martinson Honors Program
- GT Venture Capital Club, Cohort 5
- 'Terra Trends' Data Visualisation Team, *Big Data Big Impacts@GT*

Atlanta, GA

Aug. 2025 – May 2029

Normanhurst Boys High School

Valedictorian, Highest Graduating Rank in Australia

Sydney, NSW, Aus

Feb. 2019 – Oct. 2024

EXPERIENCE

Synopsis Education

Director + CTO

May 2025 – Present

Sydney, Aus (Remote)

- Engineered front-to-back tech stack for student-portal, landing website, and CRM + CI/CD automations.
- Developed full-stack web application in Typescript, React, PostgreSQL + Supabase, Docker, Express.js & Redis with REST web services and AWS S3.
- Contributed to 34k+ Git-tracked lines of production code.
- Wrote custom model context protocol in Python + FastMCP for quick powerpoint ideation.

Virtusa Polaris

Software Engineering Intern

Aug 2022

Sydney, Aus

- Learned industry-standard SCRUM techniques in the software services sector.
- Designed full-stack internal communications platform in Vue.js, GraphQL and PostgreSQL + containerized using Docker.

PROJECTS

I-IMO | AI-Powered Conversational Agent — Winner @ Calhacks 12.0

Sep 2025

- Trained a custom yolov8 model for consistent face detection and thumbnail generation with 93% face-repition accuracy.
- Used inference hosted on Groq for chunk-based text to speech streaming, synced with visual inference.
- Vectorised transcriptions in ChromaDB for high-speed semantic searches.
- Implemented profile summaries, task generation, future meeting assistance, and prompt-based context searches using RAG.

Promptly | World's 1st Prompt-Based Agent with System-Wide Control — Ergo Challenge @ HackGT25

Sep 2025

- Used Omni-Parser pre-trained weights and RICO dataset to train a lightweight, local YOLO11n object detection model with 110x faster inference times.
- Utilised OpenCV for 50ms zero-shot text detection + pyautogui for system-wide control.
- Designed a feedback-driven LLM event loop for automatic task correction.
- Speech-to-text inference using PortAudio and gpt-4o-transcribe with ~2ms inference times.
- Outperforms Gemini 2.5 Computer Use Model by 10.5x (based on inference times).

Peggiorator | Web-based, High Performance & Extensible Music Visualiser

Jan 2025 – May 2025

- Used a self-derived WASM implementation of the Fast-Fourier Transform & Web Workers to decompose input audio, 4x faster than V6 implementation.
- Captured system audio using a helper-binary written in Swift and ScreenCaptureKit.
- Injected audio data into a Vite + React sidebar component for the DOM.
- Used Zig and WebGL for 3D rendering of manipulated audio waveforms.
- Shipped a production-ready Desktop-version using Electron.

AWARDS

- **Winner**, Calhacks 12.0 @ UC Berkeley (MLH Sponsor Track – Best Use of DigitalOcean Gradient)
- **Top 0.05%** (Top 50) nationally in Australia for Board Exams (Achieved Maximum ATAR of 99.95 in HSC Board Examinations)
- **State Rank 10th** in New South Wales for Mathematics Extension 2 (Highest-level mathematics course in Australia)
- **High Distinction** in Australian Physics & Chemistry Olympiads, Silver Medal in Informatics Olympiad
- **Scientia Scholar** and **Best Mathematics Student in NSW** awards by the University of New South Wales

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), Typescript, HTML/CSS/LaTeX, R, Golang, Zig, Rust, WebAssembly

Frameworks: React, Node.js, Express.js, Next.js, FastMCP

Developer Tools: JIRA, Gradle, Git, Docker, NeoVim, AWS + any IDE you want me to learn

Libraries: pandas, NumPy, Matplotlib, Raylib, OpenGL, WebGL, torch, TensorFlow, transformers, Flask